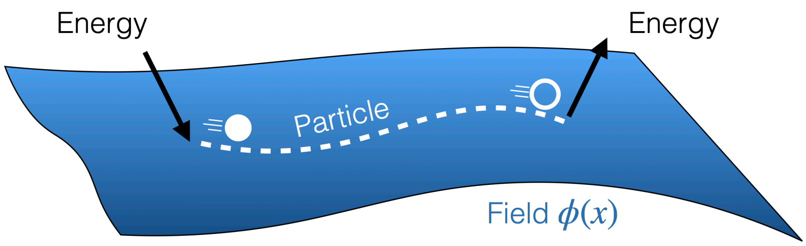
2 The Standard Model: a crash course

Before I can tell you about beyond-the-Standard Model physics, we need to define what the Standard Model (SM) actually is. To do the SM justice would require much more machinery than what most undergraduates or first-year graduate students are equipped with. However, with enough quantum mechanics and special relativity, one can get quite far, and that is precisely why these lectures exist. That being said, I have to emphasize that the SM is a quantum field theory (QFT): it describes phenomena that are both relativistic and quantum-mechanical in nature. To fully appreciate the SM in all its glory, one would need to understand some advanced concepts such as field theory, second quantization, gauge symmetries, renormalization, some cosmology, and group theory. What we are covering in these notes is then just a taste that will hopefully inspire you to take more advanced courses in these areas.

A key aspect of QFT is that particles can be created or annihilated in and out of existence: this fact is an unavoidable consequence of the principle of relativity when applied to the realm of quantum mechanics. The equivalence of mass and energy ($E = mc^2$, now more generally expressed as $E^2 = p^2c^2 + m^2c^4$) is embedded in the very formalism in QFT through the process of "second quantization". This is fundamentally different from what you may be used to in non-relativistic quantum mechanics or in special relativity classes because in those cases, the number of particles in a given system was always pre-defined and remained constant. In QFT, this is not the case; the number of particles can fluctuate and that has profound implications for how we understand particle interactions and even what the vacuum looks like (spoiler alert: it is not so empty after all).



Quantum Fields. The fundamental building blocks of a QFT are *quantum fields*. These are mathematical objects that permeate all of spacetime and can be thought of as blueprints for particle creation and annihilation. Each type of particle is associated with a specific quantum field, and the excitations of these fields correspond to the particles we observe. For example, the field $\psi(x)$ might be responsible for creating and destroying quantum states of the particle ψ, namely $|\psi\rangle$. If sufficient energy is injected into a field, it may excite a particle, and a particle may be annihilated back into the field to release energy (see above). The ways in which fields interact with each other are governed by the symmetries of the theory, and whatever interactions are allowed by these symmetries will allow excitations in one field to create or annihilate excitations in another field. In a similar way, multiple fields may participate in the creation of a single particle. Such a particle is then just a quantum superposition of excitations in different fields. This may sound like an exotic possibility, but these superpositions are ubiquitous in the SM and will be crucial to understand the physics of massive particles.

Fields can also be found in configurations that are not associated with single particle states, such as in solitons or instantons solutions. Solitons are localized objects that carry energy and are often stabilized by the topology of the field configuration or by some conserved charge, while instantons represent tunneling processes between the possible vacua in the theory. We will not discuss these further here, but they are important aspects of QFT that explain phenomena such as vacuum tunneling, topological defects (domain walls, strings, vortices, etc), and contribute to our understanding of non-perturbative phenomena in gauge theories.

> An important property of quantum fields is that the same field that creates/annihilates particles *can also create/annihilate antiparticles.* We will see exactly what types of particle and antiparticles below, but for now, it suffices to say that every particle has a corresponding antiparticle of opposite quantum numbers.

Building the Standard Model. In its most basic essence, the SM is the combination of three things:

1. a list of symmetries that determine the force carriers[1] (the spin-1 bosons: photons, W and Z bosons, and gluons),
2. a list of matter field content with their respective quantum numbers (it turns out, these are all spin-$\frac{1}{2}$ fermions, namely the quarks and leptons), and
3. a recipe for how these symmetries are broken (the spin-0 Higgs field and its potential).

This three-step process is really how most theories in high-energy physics are constructed, an art that goes by the name of *model-building.* The Higgs boson will be quite an important player in this game. Despite being the simplest mathematical ingredient in the theory (it is just a complex-valued field, as opposed to a field that evaluates to a vector or a spinor), it comes in to mess with the symmetry structure of the theory in a non-trivial way. We will not cover this mechanism in detail, but you should at least have heard that it is important.

Mediators. Each force in the SM has its own set of mediator particles (see below). In the case of electromagnetism, we have the one and only photon (γ). For the strong force, there are 8 gluons ($g_1, g_2, \ldots, g_8$), while for the weak force we have the W^+, W^-, and Z^0 (or just Z) bosons. As we will see, the photon is quite the exception: not only is it the single mediator of the electromagnetic force, but it is also the only force carrier in the SM that can mediate long-range interactions (i.e., the Coulomb force) and the only force carrier in the SM that does not interact with itself.

Each mediator is said to "couple" to a specific property of the matter particles. In the case of EM, this is just the good old electric charge. For the electron, for instance,

[1] That a symmetry should imply the existence of a particle is not at all obvious but it is true nonetheless for the *local gauge symmetries* of the SM.

Boson	**Feynman diagram**	**Mass**	**Name**
Photon (γ)	γ	$m_\gamma = 0$	Electromagnetic
$W^\pm$ boson	$W^\pm$	$m_{W^\pm} \approx 80\,\text{GeV}$	Weak (charged currents)
Z boson	Z	$m_Z \approx 91\,\text{GeV}$	Weak (neutral currents)
Gluon (g)	$g_{a\bar{b}}$	$m_g = 0$ (Confined)	Strong force ($a, b \in \{R, B, G\}$)

$Q_{em} = -1$. For the strong force, this is the more abstract color charge, schematically represented by red (R), blue (B), and green (G), and their anti-color counterparts, anti-red ($\bar{R}$), anti-blue ($\bar{B}$), and anti-green ($\bar{G}$). For the weak force, it is weak isospin T_3 and weak hypercharge Y_W. How strongly a mediator couples to a given charged particle depends not only on the charge but also on the *coupling constant*, which is the same for interactions that proceed through that particular force. For electromagnetic interactions this is just the constant $e = 0.303$ (or equivalently $\alpha_{em} = e^2/4\pi \approx 1/137$). Despite its name, the weak force also has quite a large coupling constant, $g_{\text{weak}} \simeq 0.65$. As we will see, the reason for calling the weak force has more to do with the masses of its mediators than with the strength of its interactions. For the strong force, the coupling constant is denoted by g_s. Its precise value will change by a lot depending on the energy involved in the process, but for most applications at low energies, $g_s > 1$.

The SM is elegant because all fundamental interactions are understood through the same mathematical framework of symmetries and, once all quantum numbers of the matter fields are specified (such as electric charge, color charge, spin, etc), all fundamental interactions can be uniquely determined. Not all forces are manifestations of the same symmetry, however, but they all follow the same underlying principles.

> The electromagnetic and weak forces are different manifestations of the same electroweak force. That means that γ, Z, and $W^\pm$ are all mediators of the electroweak force. At low energies, however, where classical electromagnetism is sufficient to explain our world, this unification is all but hidden by the fact that the photon is the only force carrier of the electroweak interaction that is massless. In fact, since the mass of the Z and W bosons are different, we are often required to treat EM, charged-current, and neutral-current weak interactions separately.

This kind of "unification" is not realized between the strong force and the electroweak force, at least not at energy scales we are able to probe with current experiments. Ideas of grand unified theories (GUTs) that aim to unify all three forces have generated great excitement in the past, but the most minimal models have been mostly excluded by experimental data.

Matter content. The matter fields of the SM consist of multiple generations of fermions. Those that carry color charge are called quarks, while those that do not are called leptons. They come in two types: "left-handed" (LH) and "right-handed" (RH). Only the left-handed particles participate in charged-current weak interactions, while right-handed particles do not.

> The property that distinguishes these two types of particles is called *chirality*: LH particles have *negative* chirality, while RH particles have *positive* chirality. One can define the action of the chirality operator $\hat{\mathscr{C}}$ on *chiral states* as follows:
>
> $$\hat{\mathscr{C}} \left|\psi_L\right\rangle = - \left|\psi_L\right\rangle , \qquad \hat{\mathscr{C}} \left|\psi_R\right\rangle = + \left|\psi_R\right\rangle . \tag{2.1}$$

In this sense, chirality is as much a property of the particle as its charge, spin, or any other quantum number.[2] The reason you may not have heard of this before is because chirality is often irrelevant for electromagnetism and gravity. This is because the interactions mediated by the photon and the graviton do not distinguish between left-handed and right-handed particles. This, however, is not the case for the weak interactions, where the distinction absolutely matters. Therefore, while in EM you can get away with working with the superposition

$$\left|e\right\rangle = \frac{\left|e_L\right\rangle + \left|e_R\right\rangle}{\sqrt{2}}, \tag{2.2}$$

in weak interactions, one must be more diligent and keep track of each chiral state.

One may also define projection operators,

$$\hat{P}_L = \frac{1}{2}(1 - \hat{\mathscr{C}}), \qquad \hat{P}_R = \frac{1}{2}(1 + \hat{\mathscr{C}}), \tag{2.3}$$

such that

$$\hat{P}_L \left|\psi_L\right\rangle = \left|\psi_L\right\rangle , \qquad \hat{P}_L \left|\psi_R\right\rangle = 0, \qquad \hat{P}_R \left|\psi_L\right\rangle = 0, \qquad \hat{P}_R \left|\psi_R\right\rangle = \left|\psi_R\right\rangle , \tag{2.4}$$

and

$$\hat{P}_L + \hat{P}_R = 1, \qquad \hat{P}_L \hat{P}_R = 0, \qquad \hat{P}_L{}^2 = \hat{P}_L, \qquad \hat{P}_R{}^2 = \hat{P}_R. \tag{2.5}$$

Particles of different chirality can interact with each other. But most importantly, particles of opposite chirality can mix into what we call a Dirac fermion (this is essentially what Eq. (2.2) is describing). This is the first example of a superposition of field excitations

[2] In reality, chirality may be seen as even more fundamental as it is defined by the mathematical representation of the fermion under the spacetime symmetries (namely the Poincare group: translation, rotation, and boost invariance). Weyl fermions have definite chirality and come only in two types: left-handed and right-handed. Dirac fermions, such as the electron, are a combination of both.

we alluded to earlier and it is the key mechanism by which all charged fermions in the SM, such as the electron, muon, tau, and all quarks acquire mass.

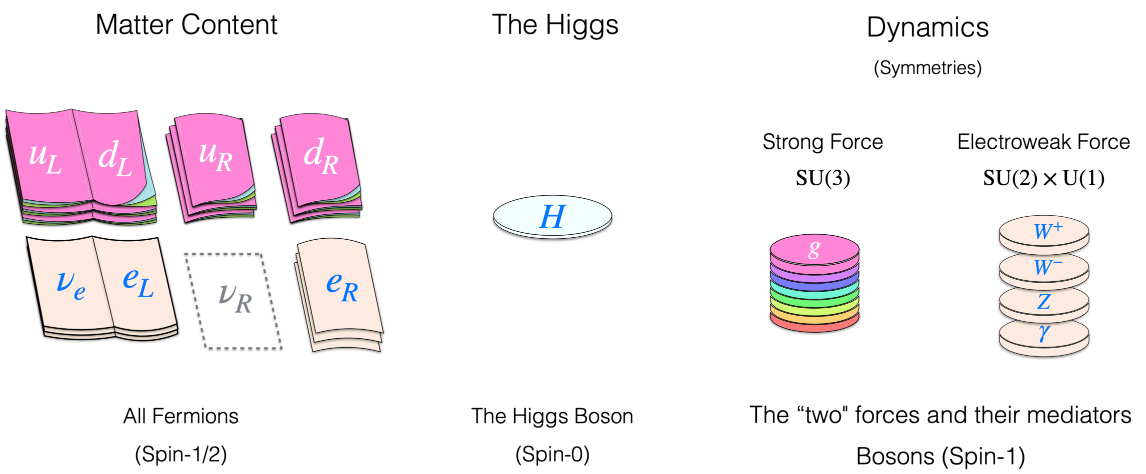
Now that we have established that particles of different chiralities are, in principle, completely different things, I can now show you the full *matter content* of the SM:

Type	Left-handed Fermions	Right-handed (RH) Fermions
Leptons	$\begin{pmatrix} \nu_L^e \\ e_L \end{pmatrix}, \begin{pmatrix} \nu_L^\mu \\ \mu_L \end{pmatrix}, \begin{pmatrix} \nu_L^\tau \\ \tau_L \end{pmatrix}$	$e_R,\ \mu_R,\ \tau_R$
Quarks	$\begin{pmatrix} u_L^i \\ d_L^i \end{pmatrix}, \begin{pmatrix} c_L^i \\ s_L^i \end{pmatrix}, \begin{pmatrix} t_L^i \\ b_L^i \end{pmatrix}$	$u_R^i,\ d_R^i,\ c_R^i,\ s_R^i,\ t_R^i,\ b_R^i$

Here, $i = R, G, B$ runs over the three color charges for quarks. Each generation consists of one pair of LH leptons and one pair of LH quarks (per color), plus one RH charged lepton and two RH quarks (per color). Counting the number of matter fields gives: 3 generations $\times$ (2 LH leptons + 2 LH quarks + 1 RH lepton + 2 RH quarks) = 3 generations $\times$ 7 fields = 21 matter fields.

Why not 24? Where did the three RH neutrinos go? Well, we did not count them because we do not actually know if they exist of not. This is a major ambiguity in the SM: do ν_R states exist or not? How many of them are there? And how do they fit into the mechanism of neutrino masses? We know that all SM fermions combine into Dirac particles, but neutrinos are special and do not necessarily have to do so. We will explore this ambiguity in more detail in the next module.

Another way to visualize the field content of the SM is the following:



In this sketch, each page or element corresponds to a specific field or particle in the SM. The three generations are represented as separate layers or sections. The different color-charged quarks are indicated by pages of different colors. And finally, the left-handed and right-handed components versions of each fermion are represented as separate piles, with the right-handed neutrino pile left purposefully empty.

Color charge. Color charge has absolutely nothing to do with what the usual concept of color in everyday life. The name "color" exists to provide an important analogy: just as there are three primary colors that come together to mix into a color-neutral combination, so too can the color charges. In an abstract sense, $RBG =$"white" and $\overline{RBG} =$"black". This is pretty much the extent of the analogy, so insist on using it at your own peril! Color neutrality is the norm in nature. Quarks and gluons are never found in isolation; they are always confined within larger composite particles with no net color charge. This is because the strong force is so strong that any net color charge is quickly neutralized by the creation of quark-antiquark pairs from the vacuum, so as to screen it. This is significant and goes by the name of *confinement.* It is as though electrons and positrons could spontaneously pop into existence to neutralize any net electric charge created. Thankfully, the electromagnetic force is neither strong enough nor complex enough to allow for this to happen. Confinement also explains why the strong force cannot be long-range: any attempt to separate two color-charged particles will result in the creation of new quark-antiquark pairs to neutralize them.

The color-neutral three-quark bound states alluded to above are referred to as *baryons.* The proton and the neutron are examples of baryons: $|p^+\rangle = |u\,u\,d\rangle$ and $|n\rangle = |u\,d\,d\rangle$. Quark-antiquark pairs can also combine into a color-neutral states, which we refer to as *mesons.* The latter are typically lighter and less stable than baryons, but are energetically much easier to produce.

Baryons ($q_a q_b q_c$ or $\bar{q}_a \bar{q}_b \bar{q}_c$):

$$\left|p^+\right\rangle = |u\,u\,d\rangle\,, \quad \left|p^-\right\rangle = \left|\bar{u}\,\bar{d}\,\bar{d}\right\rangle, \quad \left|\Lambda^0\right\rangle = |u\,d\,s\rangle\,, \quad \left|\Lambda_c^+\right\rangle = |u\,d\,c\rangle\,.$$

Mesons ($q_a \bar{q}_a$):

$$\left|\pi^+\right\rangle = \left|u\bar{d}\right\rangle, \qquad \left|\pi^0\right\rangle = \frac{|u\bar{u}\rangle - \left|d\bar{d}\right\rangle}{\sqrt{2}}, \qquad \left|K^+\right\rangle = |u\bar{s}\rangle\,.$$

We will not dedicate much time to the strong force, but you should know that its structure is much richer than the electromagnetic and weak ones. The fact that quarks and gluons are *confined* within baryons and mesons, gives rise to a whole host of new interesting dynamics. The number of composite states you can form by combining quarks together is enormous and, frankly, not fully understood in some cases. While we know the mathematical description of the strong force through quantum chromodynamics (QCD), we do not necessarily know how to perform day-to-day calculations with it because perturbation theory, our main tool in the QFT toolbox, fails. Perturbation theory is the very basis of Feynman diagrams, so even those will be of limited use when it comes to QCD.

Instead, QCD will give way to effective models and phenomenological approaches that capture the essential features of the strong force without requiring a full understanding of the underlying theory, such as Chiral Perturbation Theory. Alternatively, one can brute force their way through the problem. This is the approach of *lattice QCD*, a non-perturbative approach that discretizes spacetime and solves QCD dynamics fully numerically. This approach has been tremendously successful in relatively small systems, but it becomes increasingly difficult to apply in larger, more complex systems like nuclei.

Weak Isospin and Hypercharge. The weak force has two types of charges: weak isospin T_3, which is the only charge the $W^{\pm}$ bosons care about (the charged current), and weak hypercharge Y_W, which is the charge that the Z boson couples to (the neutral current).[3] While only left-handed particles carry weak isospin, all SM particles carry *weak hypercharge* Y_W, therefore:

> All SM fermions interact with the Z boson, but only left-handed fermions participate in weak interactions mediated by the $W^{\pm}$ bosons.

The weak isospin charge T_3 comes in two types for SM fermions: $T_3 = \pm\frac{1}{2}$ for left-chiral particles and $T_3 = 0$ for right-chiral particles. For example, ν_L^e carries charge $T_3 = +\frac{1}{2}$, e_L carries $T_3 = -\frac{1}{2}$, and e_R is neutral under weak isospin ($T_3 = 0$). This is why we say that only left-chiral particles participate in charged-current weak interactions. This assignment of weak isospin is not arbitrary. As we alluded to earlier, the left-chiral fields all come paired up; the fact they have the same but opposite weak isospin is just a reflection of this pairing. While weak isospin has nothing to do with spin, the mathematical structure of weak isospin is similar to that of spin (hence the name). The two left-chiral fields inside a given pair (like ν_L^e and e_L) are related in the same way that the two states of a spin-1/2 particle ($|\uparrow\rangle$ and $|\downarrow\rangle$) are related. In group theory terms, this just means that the left-chiral fields come in isospin SU(2) doublets, in the same way that spin-1/2 fermion is a doublet under spin SU(2).

Having introduced the full SM field content, I can now give you the full charge assignment of SM fields. This is given in Table 1, including the electric charge, weak isospin, and hypercharge.

Interactions and Feynman Diagrams. Let us look at a few examples of interactions between particles in the SM. To do so, I will introduce the concept of a Feynman diagram. If you have never seen this before, just think of it as a way to represent particle interactions visually. Each diagram corresponds to a particular possibility for a given reaction to happen. The rules of the SM determine what lines and vertices exist, and how they can be connected. Once that is specified, whatever valid diagram you draw will correspond to a real physical process. The amplitude for such a process (and its probability) are then calculable from the diagram using standard rules. The more vertices, the more suppressed the amplitude is supposed to be.

[3] To be more precise, hypercharge is a combination of weak isospin and electric charge. In reality, the Z boson couples to the combination $\cos\theta_W T_3 - \sin\theta_W Y_W/2$, where θ_W is called the Weak mixing angle.

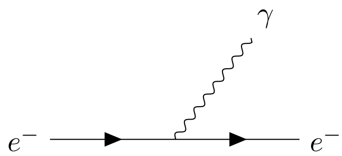
Particle	**Electric Charge** Q	**Color Charge**	**Weak Isospin** T_3	**Hypercharge** Y_W
Leptons				
ν_{eL}	0	singlet	$+\frac{1}{2}$	-1
e_L	-1	singlet	$-\frac{1}{2}$	-1
e_R	-1	singlet	0	-2
Quarks				
u_L^i	$+\frac{2}{3}$	triplet ($i = R, G, B$)	$+\frac{1}{2}$	$+\frac{1}{3}$
d_L^i	$-\frac{1}{3}$	triplet ($i = R, G, B$)	$-\frac{1}{2}$	$+\frac{1}{3}$
u_R^i	$+\frac{2}{3}$	triplet ($i = R, G, B$)	0	$+\frac{4}{3}$
d_R^i	$-\frac{1}{3}$	triplet ($i = R, G, B$)	0	$-\frac{2}{3}$
Gauge Bosons				
γ (photon)	0	singlet	0	0
g_{ij} (gluon)	0	octet	0	0
W^+	$+1$	singlet	$+1$	0
W^-	-1	singlet	-1	0
Z	0	singlet	0	0
Higgs h	0	singlet	$+\frac{1}{2}$	$+1$

Table 1: Electric charge, color charge, weak isospin, and hypercharge assignments for all SM fields (for a single generation of fermions). For quarks, "triplet" refers to the three color assignments ($i = R, G, B$); for gluons, "octet" refers to the eight gluon states that are linear combinations of $i, j = R, G, B$.

In these Feynman diagrams, we use these conventions (with examples):

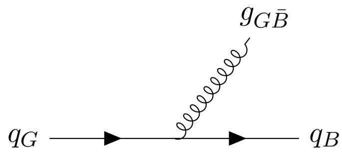
- Time flows left to right.
- Solid lines: fermions (e.g., electron e^-:)
- Wavy lines: gauge bosons (e.g., photon γ:)
- Curly lines: gluons (e.g., g:)
- Dashed lines: scalars (e.g., Higgs h: - - - -)
- Fermion arrows: for particles, for antiparticles.

First, the photon interaction with an electrically charged electron:



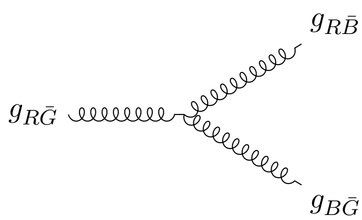
(vertex: e^- emits a photon.)

This should feel familiar. Now, consider a quark interacting with a gluon:



(vertex: green quark emits a $g_{G\bar{B}}$ gluon and becomes blue.)

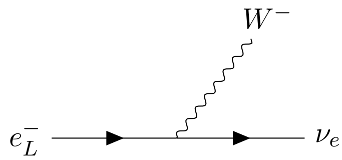
Note that color charge is still conserved. Now, because gluons carry color charge as well, they can also interact with each other. For example,



(vertex: $g_{R\bar{G}} \to g_{R\bar{B}} + g_{B\bar{G}}$.)

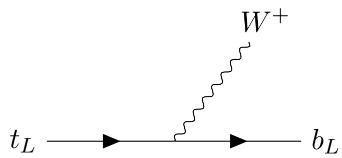
Similar four-gluon vertices exist. This self-interaction makes strong interactions even richer.

Now, weak interactions mediated by the $W^\pm$ bosons are even more interesting. Because the $W^\pm$ bosons are charged, they can change the flavor of the fermions they interact with. For example, consider the process by which an electron converts to a neutrino by emitting a W^- boson:



(vertex: $e_L^- \to \nu_e + W^-$.)

Note that only LH particles are involved. This exact process is not actually kinematically allowed ($m_e < m_{W^-}$), but for a top quark, it absolutely is:

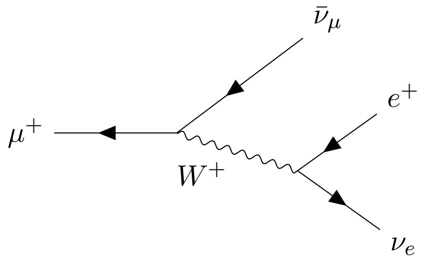


(vertex: $t_L \to b_L + W^+$.)

The process above is the dominant decay mode of the top quark. This decay mode is so fast ($\tau_t = 5 \times 10^{-25}$ s) that the top quark decays before it can even be confined inside a color neutral hadron.

This helps explain why the weak force is the "radioactive decay" force: it is the only SM interaction that can change the fermion type (or flavor) and is therefore responsible for mediating the transition from heavy particles into lighter ones. You may argue: "how is this possible? The W mass is much larger than the mass of every other SM fermion!". And you would be correct! However, the key point is that the W bosons can appear as *virtual* particles in these processes, mediating interactions without ever being produced as real, propagating degrees of freedom.

One illustrative and very important example is given by muon decay: $\mu^+ \to e^+ + \nu_e + \bar{\nu}_\mu$. The Feynman diagram is given by

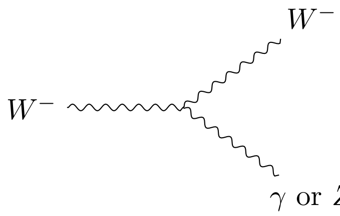


(Muon decay: $\mu^+ \to e^+ + \nu_e + \bar{\nu}_\mu$.)

Virtual particles. In this process, the W appears as a virtual particle, mediating the interaction without being directly observed. From the point of view of the muon, it only has $E_\mu = m_\mu = 105.7$ MeV in energy to give to its daughter particles. The W, being much heavier, at $M_W = 80$ GeV $\gg m_\mu$, cannot be produced, but it can still mediate the interaction provided it happens in a sufficiently short range. How short? Approximately $r \simeq 1/M_W \simeq 0.002$ fm in size. Within that range, the interaction can effectively "borrow" the mass of the W from the vacuum, allowing the decay to proceed. The cost of having this process take place in such a small region is precisely the reason why we think of the weak force as being weak.

> The usual lore behind virtual particles is that this borrowing of energy has to do with the uncertainty principle: $\Delta E \Delta t \gtrsim \hbar/2$. The more energy you want to borrow, the less time (space) you have to do so. In reality, the story is a little more complicated. We will make these ideas more precise soon, but for now, suffice it to say that muons decay due to weak interactions and that this same mechanism is what enables the beta decay of neutrons and nuclei, as well as of many of other unstable particles.

We should note that the W is electrically charged and that it also carries weak (hyper)charge. Therefore, it can interact with the photon and the Z boson,



(vertex: $W^- \to W^- + \gamma/Z$ or related triple gauge boson couplings.)

This is mostly inconsequential for our discussion here as the weak interactions will already be weak, so adding another photon or Z interaction on top of it will make it even less relevant.

Draw the leading feynman diagram for neutron decay ($n \to p^+ + e^- + \bar{\nu}_e$). Only one of the constituent quarks in the neutron should participate in this process.

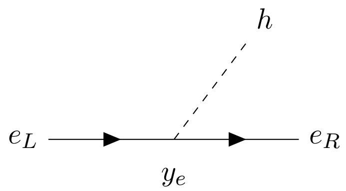
The Higgs boson. Now I have to tell you about the Higgs boson and its role in the SM. It turns out that the Higgs field has a very strange property:

It is energetically favorable for the Higgs field to take a non-zero value everywhere in space than to sit at zero, as most quantum fields do. This phenomenon gives the Higgs a vacuum expectation value (or vev), which we denote as $\langle h \rangle$. After the discovery of the Higgs in 2012, we have measured this value to be about $\langle h \rangle \approx 246$ GeV. This is the only dimensionful quantity that appears in the SM. All fermion masses are derived from it.

In the absence of interactions with other particles, the Higgs vev would be mostly inconsequential, but because all SM fermion fields (with the possible exception of the neutrino fields) interact with the Higgs, they end up developing an inherited tension from the Higgs vev that endows particles with a mass. The stronger the interaction between the fermion fields and the Higgs field is, the larger this tension is, and the larger the resulting particle masses are. In this sense, an electron interacts much less with the Higgs than the top-quark does.

These are some vague statements, but the basic idea is this: if particles are excitations of fields, then particle masses are just the minimal amount of energy that is required to excite them. You can view this as a type of tension in the fabric of the field, which can either be just an intrinsic property of the field, like spin or charge, or be inherited from interactions with other fields, like from the interaction of electron fields with the Higgs. It turns out that in the SM, no fermions carry this intrinsic mass property as they all acquire their mass entirely by interacting with the Higgs (except, maybe, RH neutrinos if they exist).

The way that particles interact with the Higgs is also special. A single chiral fermion, like e_L, cannot interact with the Higgs on its own. Instead, it has to interact with its opposite-chirality counterpart, e_R, as well as with the Higgs itself through a three-way vertex that we call a *Yukawa interaction*:



(vertex: $e_L \to e_R + h$ with y_e the Yukawa couplings between the electron and the Higgs.)

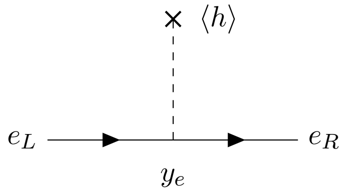
What this means is that masses are given to pairs of chiral fermions: the Dirac fermion we discussed previously. That is why all SM fermions are *Dirac fermions*, the Higgs requires two distinct chiral fields to make up a single massive state. Why? Because of weak (hyper)charge conservation. The Higgs carries weak charge and so do the LH and RH fermions. In fact, they all carry different charges. So, in order for this interaction to conserve weak charge, both chiralities must be present to balance things out.

For example, in the SM, the LH and RH electrons carry a weak hypercharge of $Y_W(e_L) = -1$ and $Y_W(e_R) = -2$, while the Higgs carried $Y_W(h) = +1$. So, no three-way combination of these fields can conserve weak hypercharge unless both chiralities are present (you can check that the Feynman diagram above conserves Y_W). While I have discussed weak hypercharge here, you can also do a similar analysis for the weak isospin T_3, which must also be conserved.

> If ν_R does exist, then it must be a completely *sterile fermion*, that is, it would carry no charge and would not participate in any interactions. You can convince yourself of this by looking at the hypercharge conservation in the interaction of the type $\nu_L + h \to \nu_R$. It must be the case that $Y_W(\nu_R) = 0$ since $Y_W(\nu_L) = -1$ and $Y_W(h) = +1$. Time permitting, we will return to neutrino masses in greater detail later to show that the existence of ν_R is not necessarily such an harmless thing to do.

Symmetry breaking? We have casually alluded to the fact that the SM Higgs "breaks electroweak symmetry". In perhaps some more familiar words, you can say that the Higgs field acts to hide charge conservation. Not electric charge (that one comes out to always be conserved and very clearly so), but weak hypercharge Y_W and weak isospin T_3. Here I chose to use the word "hide" because the way that the Higgs breaks the symmetry laws is special. They are said to be broken "spontaneously". The key here is that the symmetry is not actually broken. It just appears to be broken in the current state the Higgs field finds itself in (which is just its lowest energy configuration, often just referred to as the "ground state" or the vacuum of the theory). So, whatever charge may look to be violated will actually turn out to be conserved when accounting for how it was moved around by interactions with the Higgs field. In other words, weak charge can only be "violated" in ways that the Higgs field allows for.

One naive way to understand this hiding of weak charge is to imagine that the Higgs field can store and lend charge.[4] In the context of the interaction with the electron, this looks like the following:



$$\longrightarrow \qquad m_e = \frac{y_e \langle h \rangle}{\sqrt{2}}$$

(vertex: $e_L \to e_R$ via Higgs vev insertion. The electron mass comes out to be the product of the Yukawa coupling and the Higgs vev, up to a factor of $\sqrt{2}$.)

The cross represents the fact that the interaction has happened not with a quantum of the Higgs boson, but rather with the vev of the Higgs field, which has stored an amount of hypercharge to balance everything out. This interaction is rather curious: it transforms a LH electron into a RH electron, effectively changing its chirality.

This diagram with the cross on the Higgs field is what we usually understand as the mass term for the electron. Following our analogy of mass as the tension on the field; you can see the diagram above as imposing such a tension onto the combination of left- and right-chiral fields simultaneously. So, when an electron is created, it is the result of a superposition of left- and right-chiral states, each with their own properties. If only a Dirac mass term is present, this superposition is maximally mixing only e_L and e_R. If there were more interactions, between e_L and μ_R, for example, then more states would be involved and the massive propagating particles would look like mixtures of e_L, e_R, and μ_R. That is not the case for charged lepton, but it is the case for quarks and is the case for neutrinos as well, although the latter have other complications we will get to later.

It is also important to distinguish the diagram with the cross from the one without it. After the Higgs acquired a vev, $h(x) \to h(x) + \langle h \rangle$, these two diagrams are doing different things. When the Higgs *particle* is involved, the diagram does not represent a mass; it is simply showing the emission of a higgs from an electron line. But if the higgs background $\langle h \rangle$ is involved, then it does correspond to a mass term.

> What are the values of the lepton Yukawa couplings y_e, y_μ, y_τ? How about that of the top quark, y_t?

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[4] This is not literally true, the Higgs vev is just a constant that carries no charge. What is really happening is that the particle states we use to describe the vacuum or ground state of the theory are not the full particle states of the symmetric theory. So, while it may look like weak charge is violated by processes involving the particles we observe, if you were to track the states of the full theory, it would actually come out to be conserved. It is just that those full symmetric states are no longer useful to describe physical aspects of the theory like propagation, scattering, etc. Clearly, there are measurable and physical consequences of the symmetry being spontaneously broken, but this has more to do with the physical basis Higgs vev has forced us to work with than with the underlying symmetries themselves.